CBSE Class 10 Mathematics — Surface Areas and Volumes — Solutions

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1. **1.** If the radius of a sphere is 7 cm, then its surface area is:
   Answer: (C)
   Solution:
   1. State the formula for the surface area of a sphere: $A = 4\pi r^2$.
   2. Substitute $r = 7$ cm and $\pi = \frac{22}{7}$.
   3. Calculate the final value.

   Derivation:
   $$A = 4 \times \frac{22}{7} \times 7 \times 7 = 4 \times 22 \times 7 = 616 \text{ cm}^2$$
   Surface Areas and Volumes — Surface area of sphere

2. **2.** The volume of a right circular cylinder with radius 3 cm and height 7 cm is:
   Answer: (B)
   Solution:
   1. State the formula for the volume of a cylinder: $V = \pi r^2 h$.
   2. Substitute $r = 3$ cm, $h = 7$ cm, and $\pi = \frac{22}{7}$.
   3. Calculate the final value.

   Derivation:
   $$V = \frac{22}{7} \times 3^2 \times 7 = 22 \times 9 = 198 \text{ cm}^3$$
   Surface Areas and Volumes — Volume of cylinder

3. **3.** The total surface area of a cube whose edge is 5 cm is:
   Answer: (A)
   Solution:
   1. State the formula for the total surface area of a cube: $TSA = 6a^2$.
   2. Substitute $a = 5$ cm.
   3. Calculate the final value.

   Derivation:
   $$TSA = 6 \times (5)^2 = 6 \times 25 = 150 \text{ cm}^2$$
   Surface Areas and Volumes — Surface area of cube

4. **4.** If the slant height of a cone is 10 cm and the radius is 6 cm, its curved surface area is:
   Answer: (D)
   Solution:

1. State the formula for the curved surface area of a cone: $CSA = \pi rl$.
2. Substitute $r = 6$ cm, $l = 10$ cm.
3. Calculate the final value in terms of .

Derivation:

$$CSA = \pi \times 6 \times 10 = 60\pi \text{ cm}^2$$

Surface Areas and Volumes — Curved surface area of cone

5. 5. The lateral surface area of a cuboid of dimensions 8 cm × 6 cm × 5 cm is:

Answer: (C)

Solution:

1. State the formula for the lateral surface area of a cuboid: $LSA = 2h(l + b)$.
2. Identify $l = 8$ cm, $b = 6$ cm, $h = 5$ cm.
3. Calculate the final value.

Derivation:

$$LSA = 2 \times 5 \times (8 + 6) = 10 \times 14 = 140 \text{ cm}^2$$

Surface Areas and Volumes — Surface area of cuboid

6. 6. Assertion (A): The total surface area of a solid hemisphere of radius r is $3\pi r^2$. Reason (R): The total surface area of a sphere is $4\pi r^2$.

Answer: (B)

Solution:

1. The total surface area of a hemisphere includes the curved surface area ($2\pi r^2$) and the area of the circular base (πr^2). Thus, TSA = $3\pi r^2$. Assertion is true.
2. The formula for the total surface area of a sphere is indeed $4\pi r^2$. Reason is true.
3. The reason explains the surface area of a sphere, not the derivation of the hemisphere's area. Thus, R is not the correct explanation for A.

Derivation:

$$TSA_{hemisphere} = 2\pi r^2 + \pi r^2 = 3\pi r^2$$

Surface Areas and Volumes — Surface Area of Solids

7. 7. Assertion (A): If the radius of a cylinder is doubled and its height is halved, the volume remains the same. Reason (R): The volume of a cylinder is given by $V = \pi r^2 h$.

Answer: (D)

Solution:

1. Let initial volume be $V = \pi r^2 h$.
2. New volume $V' = \pi(2r)^2(h/2) = \pi(4r^2)(h/2) = 2\pi r^2 h = 2V$. Assertion is false.
3. The volume formula for a cylinder is correctly identified as $\pi r^2 h$. Reason is true.

Derivation:

$$V_{new} = \pi(2r)^2 \times \frac{h}{2} = 2\pi r^2 h$$

Surface Areas and Volumes — Volume of Cylinder

8. **8.** Assertion (A): The volume of a cone with radius r and height h is $\frac{1}{3}\pi r^2 h$. Reason (R): A cone is one-third of a cylinder with the same base and height.

Answer: (A)

Solution:

1. The volume of a cone is mathematically defined as $\frac{1}{3}\pi r^2 h$. Assertion is true.
2. Geometrically, a cone occupies one-third the volume of a cylinder having the same radius and height. Reason is true and explains the assertion.

Derivation:

$$V_{cone} = \frac{1}{3}V_{cylinder} = \frac{1}{3}\pi r^2 h$$

Surface Areas and Volumes — Volume of Cone

9. **9.** Assertion (A): The slant height l of a cone is given by $l = \sqrt{h^2 + r^2}$. Reason (R): This is derived from Pythagoras theorem applied to the triangle formed by height, radius, and slant height.

Answer: (A)

Solution:

1. The slant height l, radius r, and vertical height h form a right-angled triangle in a cone.
2. By Pythagoras theorem, the hypotenuse $l^2 = r^2 + h^2$, so $l = \sqrt{r^2 + h^2}$. Assertion is true.
3. The Reason correctly identifies the geometric origin of the formula. Thus, R is the correct explanation.

Derivation:

$$l^2 = r^2 + h^2$$
$$\implies l = \sqrt{r^2 + h^2}$$

Surface Areas and Volumes — Cone Properties

10. **10.** Assertion (A): If the surface area of a sphere is 616 cm^2, its radius is 7 cm. Reason (R): The surface area of a sphere is $4\pi r^2$.

Answer: (A)

Solution:

1. Using $4\pi r^2 = 616$, we have $4 \times (22/7) \times r^2 = 616$.
2. $r^2 = 616 \times 7/(4 \times 22) = 616 \times 7/88 = 7 \times 7 = 49$.
3. $r = 7$ cm. Assertion is true.
4. Reason is the correct formula used for the calculation. Thus, R is the correct explanation.

Derivation:

$$4 \times \frac{22}{7} \times r^2 = 616$$
$$\implies r^2 = 49$$
$$\implies r = 7$$

Surface Areas and Volumes — Surface Area of Sphere

11. **11.** Two cubes each of volume 64 cm^3 are joined end to end. Find the surface area of the resulting cuboid.

Answer:

$$160 \text{ cm}^2$$

Solution:

1. Determine the side of the cube using the volume formula $a^3 = 64$.
2. Identify the dimensions of the new cuboid ($l = 8, b = 4, h = 4$) and calculate the surface area using $2(lb + bh + lh)$.

Derivation:

$a = \sqrt[3]{64} = 4$ cm. New dimensions: $l = 8, b = 4, h = 4$. SA $= 2(8{\times}4{+}4{\times}4{+}8{\times}4) = 2(32{+}16{+}32) =$

Surface Areas and Volumes — Combination of solids

12. **12.** A metallic sphere of radius 4.2 cm is melted and recast into the shape of a cylinder of radius 6 cm. Find the height of the cylinder.

Answer:

$$2.744 \text{ cm}$$

Solution:

1. Equate the volume of the sphere to the volume of the cylinder: $\frac{4}{3}\pi r_1^3 = \pi r_2^2 h$.
2. Solve for h by substituting the given values.

Derivation:

$$\frac{4}{3}\pi(4.2)^3 = \pi(6)^2 h$$
$$\Rightarrow h = \frac{4 \times 74.088}{3 \times 36} = 2.744 \text{ cm.}$$

Surface Areas and Volumes — Conversion of solids

13. **13.** The slant height of a frustum of a cone is 4 cm and the perimeters of its circular ends are 18 cm and 6 cm. Find the curved surface area of the frustum.

Answer:

$$48 \text{ cm}^2$$

Solution:

1. Use the formula for curved surface area of a frustum: $CSA = \pi(r_1 + r_2)l$.
2. Since $2\pi r_1 = 18$ and $2\pi r_2 = 6$, note that $\pi(r_1 + r_2) = \frac{18+6}{2} = 12$.

Derivation:

$$\pi(r_1 + r_2) = \frac{18+6}{2} = 12.\ \text{CSA} = 12 \times 4 = 48\ \text{cm}^2.$$

Surface Areas and Volumes — Frustum of a cone

14. **14.** A solid is in the form of a right circular cone mounted on a hemisphere. If the radius of the hemisphere and the cone is 3.5 cm and the height of the cone is 4 cm, find the volume of the solid. (Use $\pi = 22/7$)

Answer:

$$154\ \text{cm}^3$$

Solution:

1. $Volume of solid = Volume of cone + Volume of hemisphere.$
2. Apply $V = \frac{1}{3}\pi r^2 h + \frac{2}{3}\pi r^3$.

Derivation:

$$V = \frac{1}{3} \times \frac{22}{7} \times (3.5)^2 \times 4 + \frac{2}{3} \times \frac{22}{7} \times (3.5)^3 = 51.33 + 89.83 \approx 154\ \text{cm}^3.$$

Surface Areas and Volumes — Combination of solids

15. **15.** The diameter of a metallic ball is 4.2 cm. What is the mass of the ball, if the density of the metal is 8.9 g/cm^3?

Answer:

$$345.39\ \text{g}$$

Solution:

1. Calculate the volume of the sphere using $V = \frac{4}{3}\pi r^3$ where $r = 2.1$ cm.
2. Calculate mass using Mass = Density × Volume.

Derivation:

$$V = \frac{4}{3} \times \frac{22}{7} \times (2.1)^3 = 38.808\ \text{cm}^3.\ \text{Mass} = 38.808 \times 8.9 = 345.3912\ \text{g}.$$

Surface Areas and Volumes — Volume of sphere

16. **16.** A metallic sphere of radius 6 cm is melted and recast into the shape of a cylinder of radius 4 cm. Find the height of the cylinder.

Answer:

$$18\ \text{cm}$$

Solution:

1. Equate the volume of the sphere to the volume of the cylinder.
2. Substitute the formula for the volume of a sphere $\frac{4}{3}\pi r^3$ and cylinder $\pi r^2 h$.
3. Solve for the height h.

Derivation:

$$\begin{aligned} \frac{4}{3}\pi(6)^3 &= \pi(4)^2h \\ \implies \frac{4}{3}\cdot 216 &= 16h \\ \implies 4\cdot 72 &= 16h \\ \implies 288 &= 16h \\ \implies h &= 18 \text{ cm} \end{aligned}$$

Surface Areas and Volumes — Conversion of solid from one shape to another

17. **17.** Two cubes each of volume 64 cm^3 are joined end to end. Find the surface area of the resulting cuboid.

Answer:

$$160 \text{ cm}^2$$

Solution:

1. Find the edge length of the cube by taking the cube root of the volume.
2. Determine the dimensions of the new cuboid (length, breadth, height).
3. Apply the surface area formula $2(lb + bh + lh)$.

Derivation:

$\implies a = 4$ cm. New dimensions: $l = 8, b = 4, h = 4$. SA $= 2(8\cdot 4 + 4\cdot 4 + 8\cdot 4) = 2(32 + 16 + 32)$

Surface Areas and Volumes — Combination of solids

18. **18.** A conical tent is 10 m high and the radius of its base is 24 m. Find the slant height of the tent and the cost of the canvas required to make the tent, if the cost of 1 m^2 canvas is Rs. 70.

Answer:

$$Slantheight = 26 \text{ m}, Cost = \text{ Rs. } 137280$$

Solution:

1. Calculate slant height $l = \sqrt{r^2 + h^2}$.
2. Calculate the curved surface area (CSA) of the cone using πrl.
3. Multiply the CSA by the given cost per square meter.

Derivation:

$l = \sqrt{24^2 + 10^2} = \sqrt{576 + 100} = 26$ m. CSA $= \frac{22}{7}\times 24\times 26 = \frac{13728}{7}$ m^2. Cost $= \frac{13728}{7}\times 70 = 13728$

Surface Areas and Volumes — Cone

19. **19.** A medicine capsule is in the shape of a cylinder with two hemispheres stuck to each of its ends. The length of the entire capsule is 14 mm and the diameter of the capsule is 5 mm. Find its surface area.

Answer:

$$220 \text{ mm}^2$$

Solution:

1. Identify the cylindrical part length by subtracting the radii of the two hemispheres from the total length.
2. Calculate the CSA of the cylinder and the CSA of the two hemispheres.
3. Sum the two areas to find the total surface area.

Derivation:

$$r = 2.5 \text{ mm}, h_{\text{cyl}} = 14-(2.5+2.5) = 9 \text{ mm. SA} = 2\pi rh+2(2\pi r^2) = 2\pi r(h+2r) = 2\times\frac{22}{7}\times2.5\times(9+5)$$

Surface Areas and Volumes — Combination of solids

20. **20.** A hemispherical tank, full of water, is emptied by a pipe at the rate of $3\frac{4}{7}$ litres per second. How much time will it take to empty half the tank, if it is 3 m in diameter? (Take $\pi = \frac{22}{7}$)

Answer:

$$16.5 \text{ minutes}$$

Solution:

1. Calculate the volume of the hemispherical tank using $\frac{2}{3}\pi r^3$.
2. Find half the volume and convert it to litres ($1 \text{ m}^3 = 1000$ L).
3. Divide by the rate of flow and convert seconds to minutes.

Derivation:

$$r = 1.5 \text{ m. Vol} = \frac{2}{3}\times\frac{22}{7}\times(1.5)^3 \approx 7.07 \text{ m}^3\text{. Half Vol} = 3.535 \text{ m}^3 = 3535 \text{ L. Rate} = \frac{25}{7} \text{ L/s. Time} =$$

Surface Areas and Volumes — Hemisphere

21. **21.** A solid toy is in the form of a hemisphere surmounted by a right circular cone. The height of the cone is 2 cm and the diameter of the base is 4 cm. If a right circular cylinder circumscribes the toy, find the difference between the volume of the cylinder and the volume of the toy. (Take $\pi = 3.14$)

Answer:

$$16.74 \text{ cm}^3$$

Solution:

1. Identify the dimensions: Cone radius $r = 2$ cm, height $h = 2$ cm. Hemisphere radius $r = 2$ cm.
2. Calculate volume of the toy: $V_{toy} = V_{cone} + V_{hemisphere} = \frac{1}{3}\pi r^2 h + \frac{2}{3}\pi r^3$.

3. Determine dimensions of the circumscribing cylinder: Radius $r = 2$ cm, Total height $H = h + r = 2 + 2 = 4$ cm.
4. Calculate volume of the cylinder: $V_{cyl} = \pi r^2 H$.
5. Calculate the difference: $V_{cyl} - V_{toy}$.

Derivation:

$$V_{toy} = \frac{1}{3}(3.14)(2^2)(2) + \frac{2}{3}(3.14)(2^3) = \frac{8.}{}$$

$$V_{cyl} = (3$$

Difference $= 50.24 - 25.12 = 25.12$ cm^3.Correction: $V_{toy} = 8.373 + 16.746 = 25.12$. Difference $= 50.$

Surface Areas and Volumes — Combination of solids

22. 22. A metallic sphere of radius 6 cm is melted and recast into the shape of a right circular cylinder of radius 6 cm. Find the height of the cylinder.

Answer:

8 cm

Solution:

1. Equate the volume of the sphere to the volume of the cylinder since the material remains constant.
2. Formula for sphere volume: $V_s = \frac{4}{3}\pi r_s^3$.
3. Formula for cylinder volume: $V_c = \pi r_c^2 h$.
4. Substitute the given values $r_s = 6$ and $r_c = 6$.
5. Solve for h.

Derivation:

$$\begin{aligned}
\frac{4}{3}\pi(6)^3 &= \pi(6)^2 h \\
\frac{4}{3} \times 216 &= 36h \\
4 \times 72 &= 36h \\
288 &= 36h \\
h &= 8 \text{ cm}
\end{aligned}$$

Surface Areas and Volumes — Conversion of solids

23. 23. A bucket is in the form of a frustum of a cone with a capacity of 12308.8 cm^3. The radii of the top and bottom circular ends are 20 cm and 12 cm respectively. Find the height of the bucket. (Use $\pi = 3.14$)

Answer:

15 cm

Solution:

1. State the formula for the volume of a frustum: $V = \frac{1}{3}\pi h(R^2 + r^2 + Rr)$.
2. Substitute given values: $V = 12308.8$, $R = 20$, $r = 12$.

3. Calculate the term $(R^2 + r^2 + Rr) = 400 + 144 + 240 = 784$.
4. Set up the equation: $12308.8 = \frac{1}{3} \times 3.14 \times h \times 784$.
5. Solve for h.

Derivation:

$$\begin{aligned} 12308.8 &= \frac{3.14 \times h \times 784}{3} \\ 36926.4 &= 2461.76h \\ h &= \frac{36926.4}{2461.76} = 15 \text{ cm} \end{aligned}$$

Surface Areas and Volumes — Frustum of a cone

24. 24. A medicine capsule is in the shape of a cylinder with two hemispheres stuck to each of its ends. The length of the entire capsule is 14 mm and the diameter of the capsule is 5 mm. Find its surface area.

Answer:

$$220 \text{ mm}^2$$

Solution:

1. Determine the dimensions: Total length = 14 mm, radius of hemisphere $r = 2.5$ mm.
2. Height of cylindrical part $h = 14 - 2.5 - 2.5 = 9$ mm.
3. Surface area of capsule = CSA of cylinder + $2\times$ CSA of hemisphere.
4. Formula: $2\pi rh + 2(2\pi r^2) = 2\pi r(h + 2r)$.
5. Calculate value using $\pi = 22/7$.

Derivation:

$$\begin{aligned} SA &= 2 \times \frac{22}{7} \times 2.5(9 + 2(2.5)) \\ SA &= 2 \times \frac{22}{7} \times 2.5(14) \\ SA &= 2 \times 22 \times 2.5 \times 2 = 220 \text{ mm}^2 \end{aligned}$$

Surface Areas and Volumes — Combination of solids